

Empirical Assessment of Image Generation and Retrieval Techniques for News Article Thumbnails

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This paper presents a systematic evaluation of image retrieval and generative techniques for creating thumbnail images for news articles. A total of 39 approaches, including human-in-the-loop techniques and fully automated workflows, are used to generate image recommendations for 8,500 news headlines. The performance of the approaches is assessed with an online user study in which participants rate the fit between a given headline and different images. We share over 210,000 image recommendations together with the user ratings from the empirical evaluation as part of an open source dataset. Finally, we present an analysis of the best-performing approaches and summarize the insights as guidelines for pairing news articles and images.

CCS Concepts: • **Computing methodologies** → **Generative and developmental approaches; Image generation; • Information systems** → *Retrieval tasks and goals*; • **Applied computing** → **Publishing; • Human-centered computing** → *User studies*.

Additional Key Words and Phrases: News Media, Generative AI, Human-Computer Interaction, Multi-Media Retrieval, User Studies

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1 Introduction

Images play an important role in the news production pipeline [4, 9]. They are used for capturing the reader's attention [8], framing stories [46, 51], increase engagement with news [22], and for affecting readers' attitudes and emotional responses [12]. Thumbnails for news articles are a key element for item click-prediction in information retrieval (IR), as shown by the fact that readers may view articles based solely on the accompanying visuals [67]. Furthermore, images can be better predictors of engagement with news than headlines or text [19]. This makes news images a critical factor in creating and evaluating news recommender systems (NRSs), as they directly

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impact user engagement and satisfaction. Despite their importance, there is a research gap in the field of NRSs regarding the selection and consideration of news images for rating/ranking news items in the recommendation process [68]. This might, in part, be due to the field's focus on offline evaluation [5, 29], where the absence of testing with human participants means data on emotional responses and changes in attitude is unavailable.

On the side of news media and publishers, there are a number of tools available to newsrooms that leverage artificial intelligence (AI) to help with the search and retrieval of fitting images [26, 37] or with the generation of news images [34, 66]. However, it is still unclear which specific image properties, e.g., subject and style selection, are relevant for a given news story and how this impacts overall engagement [22]. While previous research showed that readers prefer generated images over stock images [25, 50], no systematic assessment has yet been conducted. This results in a situation where one of the most impactful factors of NRSs, the thumbnail/news image, is not properly accounted for during the recommendation process.

Our work addresses this blind spot in NRS research by conducting an empirical user study in which we pair news articles with up to 23 retrieved and 16 generated images. By recording detailed user feedback, this setting allows us to evaluate what image properties users perceive as fitting for a given type of news item. Taken together, our work makes four main contributions. First, we outline 39 distinct image retrieval and generation pipelines for assigning images to news headlines, all released in an open source repository. Second, we present an empirical evaluation of our conducted user study on the impact of different types of image recommendations on user perception. Third, we release an open source dataset with 210,000 images and ratings for training, testing, and evaluating the next generation of image recommendation workflows. Fourth, we offer guidelines for creating retrieval and generation workflows in the form of *Lessons Learned*. The research was conducted as part of the NewsImages 2025 Challenge [28] of the 15th MediaEval Workshop.¹

We would like to preface this work by emphasizing that the methods outlined here are intended to complement the works of editors and journalists. Newsrooms are confronted with a high-stakes, high-velocity stream of incoming information. Finding suitable images to accompany published articles is challenging. The approaches presented here are not intended to be deployed unsupervised, as that would pose a risk to the validity and reliability of news. The focus is specifically on how newsrooms can leverage new technological capabilities and incorporate them into the existing news production pipeline, not replacing them. The editorial staff provides an important human-in-the-loop element for critical quality control and upholds journalistic standards.

The remainder of this paper is structured as follows. Section 2 highlights the importance and role of visuals in news and recommender systems. Furthermore, we introduce the most recent retrieval and generative models from the literature. Section 3 then outlines the retrieval and generative workflows for creating image recommendations. Section 4 describes the user study that was conducted for the evaluation, discusses the results, and outlines the open-source dataset built on the basis of the user study. Section 5 presents a summary of the lessons learned for future iterations. Section 6 discusses the limitations and potential future research directions. Finally, Section 7 concludes our paper by summarizing our contributions and insights.

2 Related Work

In this section, we first look at the role visuals play in the news production. We then contextualize their role in the domain of News Recommender Systems (NRS). After discussing images and their impact on engagement in recommendation tasks, we look at image retrieval and generation techniques for assigning visuals to news articles.

¹<https://multimediaeval.github.io/editions/2025/tasks/newsimages>

2.1 The Role of Visuals in News

Since the establishment of traditional print newspapers and the rise of modern digital news platforms, the practice of including relevant visuals with news articles has been an important part of journalism [20]. Such visuals serve various roles, perhaps the most notable being drawing readers' attention [45]. On digital news platforms and social media, for example, the displayed thumbnail plays a key role in how readers browse and choose content [32]. In certain cases, including the right visual can even promote readers' trust in the content and contribute to clearer, more effective storytelling [35]. In addition to drawing readers' attention, the visuals included as supplementary elements in news articles can be perceived as more effective in increasing newspaper subscriptions than textual content due to their immediacy and perceptual impact [11].

Visuals can provide a quick way to grasp the message in the news context; they can be more influential than textual content in eliciting emotional responses from readers, particularly when a news story contains a strong emotional category [70]. Hence, beyond their role in browsing, visuals may enhance user engagement by evoking emotional responses and making content more appealing. A deeper understanding of the article-image fit is, therefore, a crucial element when working on NRSs to better predict which content users are more likely to click.

2.2 News Recommendations

Nowadays, reading newspapers online has become part of everyday routines [47]. While news is typically curated manually by journalists and editors, especially the news articles that appear on the homepage, the role of AI-enabled NRSs in generating automatic recommendations is growing [7, 58]. NRSs can learn readers' preferences and identify articles that match their interests [55], or diversify users' reading lists [27]. Additionally, recommendations can also be provided to editors or journalists. For example, manually selecting related articles for a specific news article is a time-consuming task, and this task can be supported by recommender systems that automatically identify and recommend relevant articles to the editor or journalist [40]. While NRSs predominantly focus on creating recommendations based on textual features [31], a series of algorithms pursue a multimodal/multimedia approach that includes, e.g., leveraging image information to refine and personalize the news article selection [16, 36].

With the rise of large language models (LLMs), there are a number of new methods that leverage these new generative capabilities to recommend items [15], or to edit the article text, e.g., for language simplification [44]. While there are some preliminary works that combine multimodal recommendations and generative approaches to create new content, these are currently theoretical in nature [62] or restricted to a proof-of-concept without empirical evaluation [65].

We do see our work presented here as complementary to ongoing NRS research, as we do not propose any new recommender algorithm. Instead, our work here focused on understanding how certain image properties affect user perception. This information can then be leveraged in a subsequent step to inform how NRSs should select/generate/integrate visual, e.g., to increase engagement with news or user satisfaction.

2.3 Image Retrieval Techniques

Image retrieval is part of the field of IR [41], which aims to provide techniques for efficient access to information items relevant to users' needs and interests [1, 57]. In the context of images, retrieval techniques primarily focus on finding relevant image matches from a collection that best correspond to a given query or information need [13] (i.e., matches in which the image content and the input query are highly similar). Image retrieval is related to a wide range of applications, such as image captioning for textual descriptions of images, and summarization [23].

Initial approaches primarily relied on traditional machine learning models [52], while later work shifted toward deep learning methods, such as convolutional neural networks, which demonstrated substantial performance improvements [30]. More recent approaches, such as ConVIRT, have explored transformer-based models to learn cross-modal representations from text and images [71]. In *Learning Transferable Visual Models From Natural Language Supervision*, the authors further advanced this direction by proposing CLIP (Contrastive Language-Image Pre-Training), designed to improve efficiency and scalability in model learning [53]. These retrieval techniques can enable editors and journalists to browse and access existing images to assign visuals to news stories.

2.4 Image Generation Techniques

Image generation has recently emerged as a new field and has rapidly advanced over the last few years, enabling the creation of fully synthetic images, e.g., from text prompts [18]. Earlier approaches in this field relied on auto-regressive transformers, which have been successful in natural language processing tasks [64]. Hence, these models were later employed for image synthesis as well by learning high-quality image representations and enabling pixel-level prediction [10]. Subsequently, generative adversarial networks (GANs) emerged as another class of generative models [14]. These models aim to learn the probability distribution underlying the training data and use it to generate new, fully synthetic images [24]. More recently, the combination of LLMs and vision encoders for describing image content has led to the creation of vision-language models (VLMs). These VLMs can perturb the image data with additional noise and are then trained to reverse the noising procedure for a robust image generation [69]. VLMs, such as diffusion models, have been shown to be a powerful technique for generating high-resolution images [56]. It is worth noting, however, that despite their superior performance, these models still incur high computational costs and raise a range of legal and ethical concerns regarding their use [6].

In the context of news, generative approaches can help to address a series of inherent limitations of retrieval techniques. For example, while image retrieval may fall short when suitable visuals are unavailable in the database or not fully aligned with an article's context, image generation can be an alternative, as fully synthetic visuals offer a better match to an article's body/main text. However, this introduces new challenges, e.g., the credibility of the generated images and the potential risks of misrepresentation. This underscores the need for ethical guidelines when using such techniques and highlights the system designer's responsibility [63].

3 Methods

This section outlines 39 distinct retrieval and generation approaches for creating image recommendations. Section 3.1 first explains all retrieval approaches and Section 3.2 then lists the different generation workflows. The code for each workflow is available on GitHub.²

3.1 Retrieval Approaches

Overall, 22 distinct retrieval workflow configurations are used to generate image recommendations from headlines. There are five distinct groups of approaches:

- (1) runs that use a default out-of-the-box model with default workflows,
- (2) runs with an extended post-processing stage for altering or judging the model output,
- (3) runs with an extended pre-processing stage for query formulation,
- (4) runs with multi-stage extensions for pre- and post-processing steps, and
- (5) fine-tuned models.

²GitHub workflow repository: <https://github.com/Informfully/Challenges/tree/main/newsimages25/workflows>

All five groups will be outlined in greater detail in the upcoming parts. Assessing runs across these categories provides a systematic view of the impact of specific changes throughout the retrieval pipeline. Figure 2 shows an overview of the different stages involved here. Table 1 then presents an overview of all workflow configurations.

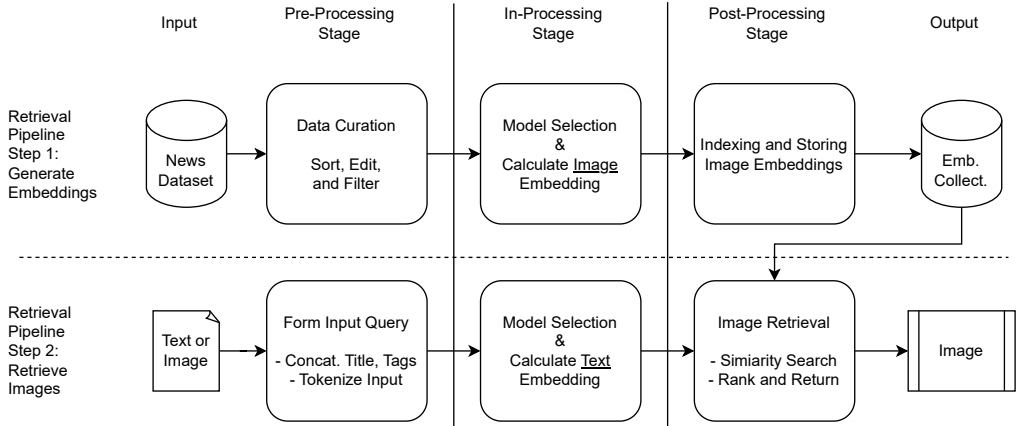


Fig. 1. Workflow overview for image retrieval. The overall process consists of two steps. Step 1 uses news and image datasets, which are processed by a model to produce a collection of embeddings for retrieval. Step 2 then uses the articles, calculates embeddings, and compares them with those in the collection. The output is the image with the most similar embeddings to the text.

Default Retrieval Workflows. The default workflow uses CLIP Base (CLIP ViT-B/32)³ for calculating text and image embeddings, together with FAISS⁴ for indexing and storing of the embeddings. The input query to the tokenizer follows a standardized template. It includes the following elements: “<General instructions>, <title>, <keywords>, <body>.”⁵

Image retrieval is performed using cosine similarity between image and text embeddings. The *Yahoo-Flickr Creative Commons 100 Million*⁶ (YFCC100M) and English articles from the *GDEL Project* (GDEL)⁷ are used as the primary dataset for image retrieval.

SEEK [34] (run ID 1-11) uses the default workflow with the out-of-the-box CLIP Base for calculating embeddings, FAISS for indexing and storing, and YFCC100M for retrieving images. It uses all available article properties to formulate the input prompt.

ITI-5 [21] (run IDs 1-05) is identical in its configuration to the previous approach. The one difference is that it uses only the article title as input for the retrieval stage.

CLIP300K [42] (run IDs 1-08 and 2-03) presents a variation of the default workflow outline above that retrieves from a subset of 300,000 images of the YFCC100M dataset. The idea behind this approach is to see how performance scales with the number of available images.

GDEL [49] (run IDs 1-21 and 2-10) indexes 8,500 GDEL images, using titles and tags as prompts. This approach serves as a baseline for performance comparison.

³<https://huggingface.co/openai/clip-vit-base-patch32>

⁴<https://github.com/facebookresearch/faiss>

⁵Tags and body are optional. The standard approach uses as large an input as possible with the model. It truncates the text if the maximum input length is reached. The general instructions read “A news photo about [...]” or variations thereof.

⁶YFCC100M website: <https://www.multimediacommons.org>

⁷<https://www.gdelproject.org/>

Table 1. Overview of the run configurations for image retrievals. Each run is listed with its ID(s), model used, and dataset. Input specifies whether the title (T), article body (B), tags/keywords (K), or image (I) are used in the prompt. Pre lists pre-processing operations, Post lists post-processing operations.

Name	Run	Model	Dataset	Prompt	Pre	Post
SEEK [34]	1-11	CLIP	YFCC	T, K		
ITI-5 [21]	1-05	CLIP	YFCC	T		
CLIP300K [42]	1-08, 2-03	CLIP	YFCC300K	T, K		
GDELT [49]	1-21, 2-10	CLIP	GDELT	T, K		
GDELT [39]	1-19	CLIP	GDELT	T, B, K		
CLIPLEX [3]	1-23, 2-18	CLIP bigG	YFCC	T, I		
ITI-1 [21]	1-01	Ensemble	YFCC	T		Judge
ITI-2 [21]	1-02	Ensemble	YFCC	I		Judge
ArtRanker [33]	1-25	Ensemble	YFCC	T, K		Ranker
VLMJudge [33]	1-27	Ensemble	YFCC	T, K		Judge
Rerank [54]	1-31, 2-01	CLIP bigG	GDELT	T, K		Ranker
CLIPSEL [3]	1-24, 2-19	CLIP bigG	YFCC	T, I		Filter
LLMCAP [33]	1-26	Ensemble	YFCC1.5M	T	LLM	
KEYBERT [61]	1-28, 2-20	CLIP	YFCC300K	T, K	NER	
Regular [2]	1-13, 2-26	Long-CLIP	YFCC50M	T	LAM	
Longtext [2]	1-14, 2-25	Long-CLIP	YFCC50M	T, B	LAM	
BLIP [17]	1-22, 2-17	CLIP	YFCC	T, I	VLM	Ranker
ITI-3 [21]	1-03	Ensemble	YFCC	B	LLM	Judge
ITI-4 [21]	1-04	Ensemble	YFCC	I	LLM	Judge
ITI-6 [21]	1-06	Ensemble	YFCC	T, I	LLM	Judge
Tuned [2]	1-18, 2-27	Long-CLIP	YFCC50M, N24	T, B	LAM	
TRACE [34]	1-12, 2-05	CLIP	YFCC, GDELT	T, K		

GDELT [39] (run ID 1-19) also indexes only GDELT. In contrast to the previous method, however, it uses all available metadata of the news article to calculate embeddings.

Exhaustive [17] (run IDs 1-23 and 2-18) uses CLIP bigG (CLIP-ViT-bigG)⁸ and uses both the article title and the original image for the retrieval process.

Post-Processing Stage Extension. A second group of approaches extends the basic workflow by adding additional steps to process the model output. The three options highlighted in this context are image rankers, judges, and filters. Both rankers and judges receive a list of images from one or more models. They then need to determine which image to select and return. Rankers follow an explicit formula set by the developer to score image properties and determine the best one. Runs with judges do not follow an explicit rule. Instead, they do so more implicitly by using an LLM or VLM to determine which image to select and return. Filters apply heuristics to remove undesired images.⁹ This also narrows down the search space and speeds up the retrieval

ITI-1 [21] (run ID 1-01) uses a family of VLMs to calculate different embeddings for presenting multiple images to the judge. This includes CLIP Base, BLIP Base,¹⁰ BLIP-2,¹¹ SLIP,¹² and Beit-3.¹³

⁸<https://huggingface.co/laion/CLIP-ViT-bigG-14-laion2B-39B-b160k>

⁹This includes, for example, images that violate the ethics policy.

¹⁰<https://huggingface.co/Salesforce/blip-image-captioning-base>

¹¹<https://huggingface.co/Salesforce/blip2-opt-2.7b>

¹²<https://github.com/facebookresearch/slip>

¹³https://huggingface.co/timm/beit3_base_patch16_224.pt

The judge in the post-processing step is Qwen 2.5¹⁴ and assesses the semantic alignment of the text and each of the five retrieved images. *ITI-2* (run ID 1-02) presents a slight variation of this, where the original news thumbnail is used as input instead of the article text.

ArtRanker (run ID 1-25) also uses a family of models to produce different image outputs for the judge. The models used here are CLIP Base, CLIP Large (CLIP ViT-L/14),¹⁵ and LargeSigLIP-2.¹⁶ These images are ranked by LAION Aesthetics Predictor v1,¹⁷ by assessing the visual quality. *VLMJudge* [33] (run ID 1-27) is a variation of this approach where the ranker in the end is replaced by a Qwen 2.5 judge. Selection criteria for determining the best picture include visual quality, trustworthiness, non-promotion of stereotypes, and non-clickbait.

Rerank [54] (run IDs 1-31 and 2-01) is an alternative extension that uses CLIP bigG to retrieve five images. These images, together with the title, are then passed through the same or a complementary model to obtain more precise embedding scores.

Selective [17] (run IDs 1-24 and 2-19) uses CLIP bigG as the model, with the title and image as inputs. It adds a filter to the post-processing stage to narrow down the search space for quicker retrieval. A secondary function of this filter is to apply business logic, e.g., filtering out images that do not comply with ethical standards.

Pre-Processing Stage Extension. Extensions to the pre-processing stage focus on creating a richer input prompt. The three methods highlighted here include LLM-based (re-)writing of the input prompt, name-entity recognition (NER), and predicting categories (LAM).

LLMCAP [33] (run ID 1-26) uses Llama 3 (Llama-3-8B)¹⁸ to generate five potential image captions based on the title of the news article. Embeddings are calculated with CLIP Base, CLIP Large, and LargeSigLIP-2. The workflow retrieves the image that is most similar to the set of original titles and all generated captions from a subset of 1.5M YFCC images.

KEYBERT [61] (run IDs 1-28 and 2-20) uses spaCy¹⁹ for NER and KeyBERT²⁰ for keyword extraction to enrich the input prompt and retrieve from a subset of 300,000 YFCC images.

Regular (run ID 1-14 and 2-26) used the Long-CLIP²¹ model with LAM (Learnable Alignment Module) [59] in the pre-processing step to learn a news category alignment based on headlines and add this to the input prompt. It retrieves from a subset of 50M YFCC items. *Longtext* [2] (run ID 1-13 and 2-25) is a variation of the previous approach that uses the title, categories, and article body (up to 248 tokens in total) as the prompt.

Combined Pre- and Post-Processing Stage Extension. This set of approaches presents multi-stage extensions that combine prompt enrichment in the pre-processing stage, together with a re-ranking strategy in the post-processing stage of the retrieval workflow.

BLIP [17] (run IDs 1-22 and 2-17) extends the pre-processing step by creating an image caption of the original article with BLIP Base. This text, along with the title and body, is used as input to CLIP Base to calculate embeddings. The post-processing step is extended by a ranker that assesses the CLIP output. It gets an embedding for the title, body, and caption. It has learned weights for how much each component contributes to the cosine similarity calculation.

¹⁴<https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

¹⁵<https://huggingface.co/openai/clip-vit-large-patch14>

¹⁶<https://huggingface.co/google/siglip2-so400m-patch14-384>

¹⁷<https://github.com/LAION-AI/aesthetic-predictor>

¹⁸<https://huggingface.co/meta-llama/Meta-Llama-3-8B>

¹⁹<https://spacy.io/models/en>

²⁰<https://github.com/MaartenGr/KeyBERT>

²¹<https://github.com/beichenzbc/Long-CLIP>

ITI-3 [21] (run ID 1-03) features an ensemble approach for embedding calculation, using CLIP Base, BLIP Base, BLIP-2, SLIP, and BeiT-3. The pre-processing stage is extended with Llama 3.2 (Llama-3.2-1B-Instruct)²² that generated a summary of the article body. This summary is then used as input to the ensemble. The ensemble output is used to retrieve five images. They are then presented to Qwen 2.5-based judge to determine the best overall candidate. *ITI-4* (run ID 1-04) presents an alternative approach in which the LLM generates a caption for the original image. It then uses this caption as an input to the retrieval model and follows the same post-processing steps as outlined before. *ITI-6* (run ID 1-06) is a special approach that first executed both *ITI-1* and *ITI-4*. It received two images, one from each approach, forwarding them to a Qwen 2.5 judge for determining the best fit with respect to the original article title and image.

Fine-Tuned CLIP Model. The process of model fine-tuning involved using additional data to update the weights of the original model, e.g., to meet the specific needs of the news domain.

Tuned [2] (run ID 1-18 and 2-27) presents one such approach, in which Long-CLIP was fine-tuned on 210,000 news articles from the N24 dataset.²³ Furthermore, it has a LAM pre-processing extension for news category identification and uses a subset of 50M YFCC items.

TRACE [34] (run IDs 1-12 and 2-05) uses a similar approach where CLIP Base is fine-tuned on GDELT using the AdamW optimizer.²⁴ Article title and tags are used as input, and images are retrieved from YFCC.

3.2 Generation Approaches

A total of 16 distinct workflow configurations are used in the generation-based approach to news image recommendations. There are four distinct groups of approaches:

- (1) runs that use a default workflow configuration for each model,
- (2) runs that vary the art style of the image,
- (3) iterative approaches that add a post-processing step, and
- (4) fine-tuned models.

The following parts will outline each of the four groups in greater detail. Figure 2 provides an overview of the generation workflow. Table 2 presents an overview of the configuration of all runs for a systematic evaluation of the impact of the model selection, title/body/keywords/negative input prompts, and post-processing operations.

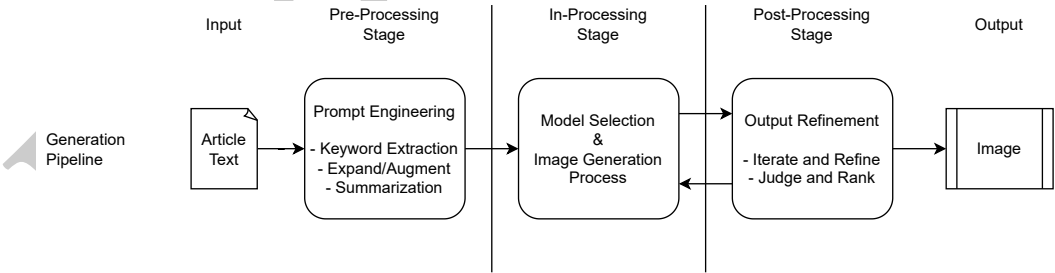


Fig. 2. Workflow overview for image generation. The overall process consists of three stages. It starts with engineering an input prompt, which is then forwarded to the model. The model output is either used as the final results or fed back into the model for subsequent refinement steps.

²²<https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct>

²³<https://www.kaggle.com/datasets/ritabrata123/n24news-zip>

²⁴<https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html>

Table 2. Overview of the run configurations for image generation. Each run is listed with its ID(s) and model. Input specifies whether the title (T), article body (B), tags/keywords (K), or negative prompts (N) are used in the prompt. Pre lists pre-processing operations, Post lists post-processing operations

Name	Run	Model	Prompt	Pre	Post
SDBASE [39]	1-20	SD 3.5	T, K, N		
SDXLBASE [66]	1-13, 2-06	SDXL 1.0	T		
SDXLTAGS [39]	1-19, 2-09	SDXL 1.0	T, K, N		
SDTURBO [42]	1-07, 2-02	SDXL Turbo	T, K, N		
SDCOLLAGE [49]	1-40, 2-23	SD 1.5	T, K, N		Collage
SDXLNEG [66]	1-14, 2-07	SDXL 1.0	T, N		Refiner
SDXLREF [66]	1-15, 2-08	SDXL 1.0	T		Refiner
LMMSummary [48]	1-35, 2-13	Infinity	T, B, K	LLM	Judge
LMMTitle [48]	1-36, 2-14	Infinity	T	LLM	Judge
HumanSelect [48]	1-34	Infinity	T, B, K		Judge
CartoonSummary [48]	1-32, 2-11	Infinity	T, B, K	LLM	Judge
CartoonTitle [48]	1-37, 2-15	Infinity	T		
RealSummary [48]	1-33, 2-12	Infinity	T, B, K	LLM	Judge
RealTitle [48]	1-38, 2-16	Infinity	T		
PromptForge [34]	1-09, 2-04	Juggernaut XI v11	T, B, K, N	LLM	Judge
VIVID [34]	1-10	Juggernaut XI v11	T, N		

Default Generation Workflows. The first family of approaches for image generation uses the basic workflows and prompting instructions provided by each VLM. The input prompt follows the schema of “<General instructions>, <title>, <keywords>, <image specifications>, <negative prompts>.” In this context, general instructions include commands like “Editorial news photo illustrating [...]” The title and tag information are read from the dataset. Image specifications include prompts such as “photo journalism, high details” to describe the image’s quality. Negative prompts, e.g., “no text, no watermark,” are model-specific and added to the end of the input.

SDBASE [39] (run ID 1-20) uses Stable Diffusion (SD) 3.5²⁵ together with the default prompting template, using as many words of the article title as possible as input.

SDXLBASE [66] (run IDs 1-13 and 2-06) follows again the default workflow and prompting template, but instead uses Stable Diffusion XL (SDXL) 1.0²⁶ for image generation.

SDXLTAGS [39] (run IDs 1-19 and 2-09) uses the same SDXL 1.0 base model and—thanks to the larger context window offered by the model—includes all article tags.

SDTURBO [42] (run IDs 1-07 and 2-09) uses the SD Turbo²⁷ variant to look at the performance impact of using lightweight models over the base and XL versions.

Image Style Variations. Cartoons (e.g., political cartoons for satire) have a long tradition in news media [60]. With the freedom that image generators offer, the art style itself becomes variable.

To assess the impact of the art style, *CartoonSummary* (run IDs 1-32 and 2-11) together with *CartoonTitle* (run IDs 1-37 and 2-15) focus on non-photorealistic images. They systematically compare them to real ones provided by *RealSummary* (run IDs 1-33 and 2-12) and *RealTitle* (run IDs 1-38 and 2-16). The summary-variants of the runs use Llama 3 for prompt generation based on the body and article tags; the title-variants only the article heading (see [48] for more details).

²⁵<https://huggingface.co/stabilityai/stable-diffusion-3.5-large>

²⁶<https://huggingface.co/stabilityai/stable-diffusion-xl-base-1.0>

²⁷<https://huggingface.co/stabilityai/sd-xl-turbo>

Iterative Generation Workflow. An iterative generation workflow is one in which the model output is fed into an additional processing step to improve the final results. Three extensions are examined in greater detail: image collages, refinement, and image ranking judges.

SDCOLLAGE [49] (run IDs 1-40 and 2-23) uses the base model of SD 1.5,²⁸ taking into account all the information shared in the dataset. For each article, this approach generates four images and combines them into a collage during post-processing.

SDXLNEG [66] (run IDs 1-14 and 2-07) uses SDXL together with SDXL Refiner²⁹ in a post-processing step. The goal of this process is to create an image with fewer flaws, e.g., visual artifacts or missing fingers, and to iteratively refine the model's output and combine it with negative prompts. *SDXLREF* (run IDs 1-15 and 2-08) is an alternative without negative prompts.

LLMSummary (runs 1-15 and 2-13) considers both the title and the article summary generated during pre-processing with Llama 3. Infinity (Infinity-8B)³⁰ is then used to generate four images (a cartoon, modern, abstract, and modern art version). The post-processing stage adds a multi-layered evaluation, where Llama 3 acts as a judge to find the image that best aligns with the article. *LLMTitle* (runs 1-36 and 2-14) is a variant of this that considers only the title as input. *HumanSelect* (run ID 1-34) presents variations of the previous two approaches in which the judge in the post-processing stage is replaced by a human (see [48] for more details).³¹

Fine-Tuned Diffusion Models. At the time the experiment was conducted, the SD-model family had already been in use for more than two years. Given the rapid progress in the field of VLMs, Juggernaut XI v11,³² an updated and fine-tuned version of SDXL was used. This allows assessing how the continued development of generation capabilities impacts performance.

PromptForge [34] (run IDs 1-09 and 2-04) presents an approach that uses this new model. The input is a prompt template created with KeyBERT for keyword extraction and negative prompt assignment (based on the article, body, and tags). *VIVID* (run ID 1-10) is identical to the former approach, but it focuses only on article titles for creating the input prompt.

4 User Study

This section provides an overview of the user study conducted to assess the performance of the retrieval and generation methods outlined in Section 3. To that end, Section 4.1 first provides a brief description of the user study's experimental setup and its subtasks. Next, Section 4.2 and Section 4.3 present the results and summary of the performance evaluation. Finally, Section 4.4 provides an overview of the collected image recommendations together with the survey results.³³

4.1 Experimental Setup

The evaluation of the retrieval and generation approaches used 8,500 English-language news articles sourced from the open-source GDELT dataset, collected between 2022 and 2023 (for details on the dataset curation, please see [38]). Each article record contained an internal article identifier, the article title (which in some cases included a lead sentence), automatically generated topical tags derived from the article body, a URL pointing to the original article, and both an identifier and URL for the editorially assigned thumbnail image. Table 3 provides an overview of the available article attributes contained within the dataset.

²⁸<https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5>

²⁹<https://huggingface.co/stabilityai/stable-diffusion-xl-refiner-1.0>

³⁰<https://huggingface.co/FoundationVision/Infinity>

³¹*HumanSelect* presents a human-AI collaborative approach for image generation. The images shown are from both *LLMSummary* and *LLMTitle*, for a total of eight images. It is then up to the human judge to select the best-fitting image.

³²<https://huggingface.co/RunDiffusion/Juggernaut-XI-v11>, <https://www.rundiffusion.com/juggernaut-xl>

³³Dataset available on GitHub:<https://github.com/Informfully/Challenges/tree/main/newsimages25/images>

Table 3. Overview of attributes for GDELT articles used for the recommendation process.

Attribute	Description
article_id	ID of news article.
article_url	Original URL of the news article.
article_title	Title of the news article (may include lead).
article_tags	Automatically generated tags for the main article text/body.
image_id	ID of news image (we provide a copy of the image).
image_url	Original URL of the news image.

A subset of 50 articles from this pool was used in an online study to assess the performance of the image recommender methods. The panel consisted of $N = 30$ English-speaking participants³⁴ and was conducted on Qualtrics³⁵. For each article, they were shown the title of a news item, along with up to 40 different image recommendations. Their task was then to assess the fit between the article and the image. The instruction shown to them read: “Please rate how fitting the image is for the given article title.” Ratings were given on a 5-point Likert scale, ranging from 1 (“Very poor fit”) to 5 (“Very good fit”).³⁶ The order of articles was randomized, as was the order of images for each article. All submitted images had the same resolution of 460×260 pixels in landscape orientation.³⁷

The online user study consisted of two subtasks. **Subtask 1** focuses on human-AI collaboration. The intent behind this task was to design a challenge that can accommodate human-in-the-loop workflows in which a journalist or editor actively guides the retrieval or generation process. To that end, 30 article IDs were *communicated in advance* so the recommendations could be manually adjusted. **Subtask 2**, on the other hand, emphasizes fully automated, scalable solutions. The remaining 20 random articles from the overall article pool were selected *without sharing* the relevant IDs beforehand. This required processing the entire collection of 8,500 articles, i.e., a number large enough to ensure that no manual curation of image recommendations was possible.

The image fit rating for a given article-image pair is the average across all users. Similarly, the score assigned to a retrieval/generation method is the average rating it received for all of its image recommendations. If a method skips or submits an invalid image for an article, a default rating of 1 is assigned. The detailed results for both subtasks are shown in the next two subsections.

4.2 Subtask 1

For subtask 1, a total of 40 methods have been tested (including the baseline). 24 approaches use retrieval techniques, 15 generative runs, and one editorial baseline. Table 4 shows the top 10 approaches for subtask 1 that cover the human-AI collaborative setting (including the baseline).³⁸ The results show that both retrieval-based and generative algorithms can produce high-quality images. Generative methods tend to outperform retrieval-based methods; the average image-fit score across the best runs for each team is higher for generative methods than for retrieval-based methods. At the same time, the evaluation shows that using editorial images presents a very competitive baseline; only six methods outperformed the baseline.

³⁴The online panel consists of experts/team members who took part in the online challenge, as well as the event organizers.

³⁵<https://www.qualtrics.com/strategy/research/survey-software/>

³⁶Figure 3 in the Appendix shows a screenshot of how the image recommendations were presented to users, together with the 5-point Likert scale to assess the image fit.

³⁷For a complete overview of the technical details on the image formats, please see the online README: <https://informfully.com/docs/recommendations.html>

³⁸Please see Table 7 in the Appendix for a complete overview of all results.

Table 4. The top 10 runs for subtask 1 (“human-AI collaboration,” incl. baseline) sorted by the average obtained article-image recommendation score (highest one on the top). Type “RET” marks retrieval-based algorithms; “GEN” the generation-based approaches.

ID	Description	Type	Score
1-12	VIVID	GEN	3.401
1-09	PromptForge	GEN	3.233
1-34	HumanSelect	GEN	3.115
1-22	BLIP	RET	3.096
1-11	TRACE	RET	3.080
1-10	SEEK	RET	3.074
1-30	BASELINE	-	3.041
1-38	TitleRealistic	GEN	3.019
1-14	Regular	RET	3.014
1-31	Rerank	RET	3.008
1-13	Longtext	RET	3.006

Subtask 1 – Retrieval Results. There is little variation in retrieval, as groups tend to use the same setup; the biggest impact is the size of the selected dataset. All top teams use the YCCC100M dataset. The three best retrieval-based methods beat the baseline, but perform worse than the best generative-based approaches. These three retrieval-based approaches use the CLIP model but differ in their configurations regarding the included pre- and post-processing steps.

The best retrieval-based approach, BLIP, applies VLM-based preprocessing and ranked-based post-processing. In addition to the news article title, an image caption for the original image in the news article is computed. The second- and third-ranked retrieval-based approaches, TRACE and SEEK, generate keywords for retrieval and do not apply post-processing. This shows that, using a standard keyword-extraction workflow, high retrieval performance can be achieved. For image retrieval, the relevant information is mainly in the news title. The computation of good keywords is important for high-quality retrieval results.

Subtask 1 – Generative Results. The evaluation shows that the overall best scores are obtained by Generative approaches using the *Juggernaut XI v11* model (used by the teams *PromptForge* and *VIVID*). This demonstrates that the fine-tuned Juggernaut models outperform older models. All models based on the standard Stable Diffusion approach perform worse than the baseline. The *VIVID* method achieved the best performance. This shows that the approach’s success is based on the fine-tuned generation model *Juggernaut XI v11*.

The second-ranked method, *PromptForge*, uses the same image generation model but adds post-processing and pre-processing that consider not only the news article title but also the body and keywords. The additional effort seems to slightly reduce the evaluation performance.

The method *HumanSelect* is ranked third. This method uses the Infinity model to generate suggestions, which a human then selects to find the best-matching image. It seems that human selection yields better performance than automated post-processing by an LLM or VLM.

Subtask 1 – Summary. Overall, both generative and retrieval-based methods reach high-quality results, outperforming the baseline. Since subtask 1 is focused on a human-in-the-loop setting, the human’s involvement also affects the results. Thus, the effort humans put into post-processing suggestions can also influence the results.

4.3 Subtask 2

Overall, 26 methods have been tested for subtask 2. 15 approaches use retrieval techniques, 11 approaches generate images, and the editorial baseline. Table 5 shows the top 10 approaches to subtask 2 for the fully automated recommendation approach, including the baseline.³⁹ As with the first subtask, we see that the editorial baseline is very competitive, and only six recommendation procedures can outperform it.

The best-performing method is *PromptForge*, which uses the Juggernaut XI v11 model to generate images. The method uses LLM-based pre-processing and judge-based post-processing during image generation. The fact that the best teams in subtask 1 are also based on the Juggernaut model demonstrates the strength of this image generation model in the analyzed scenario. Post-processing image suggestions help handle variations in the generated images. In the human-in-the-loop setting (subtask 1), manual post-processing is successful; however, the best evaluation score in subtask 2 is achieved with an automated approach.

Table 5. The top 10 runs for Subtask 2 (“fully automated pipelines,” incl. baseline) sorted by the average obtained article-image recommendation score (highest one on the top). Type “RET” marks retrieval-based algorithms; “GEN” the generation-based approaches.

ID	Description	Type	Score
2-04	PromptForge	GEN	3.194
2-05	TRACE	RET	3.114
2-17	BLIP	RET	3.088
2-06	SDXL	GEN	3.031
2-08	SDXLREF	GEN	3.024
2-14	LLMCriticTitle	GEN	2.983
2-22	BASELINE	-	2.956
2-25	Longtext	RET	2.953
2-07	SDXLNEG	GEN	2.938
2-15	TitleCartoon	GEN	2.927
2-13	LLMCritic	GEN	2.923

Subtask 2 – Retrieval Results. The TRACE approach is the best performing retrieval-based method in subtask 2. This method uses the CLIP model on the YFCC100M and GDELT datasets. TRACE involves pre- and post-processing; the news headline and keywords are used as the basis for the retrieval process. The BLIP approach is the second-best retrieval-based approach (ranked third in the subtask 2 leaderboard). This method uses news headlines and the original image for retrieving image suggestions from the YFCC100M dataset.

Subtask 2 – Generative Results. The second- and third-most effective generative methods use a Stable Diffusion model. These approaches have in common that they focus on the headline of the news article and do not consider the body of the article or negative prompts. This demonstrates that the title contains all the important information and that the image should primarily utilize cues from the headline to create image recommendations. Results indicate that leveraging more information from the body, for example, adds unnecessary noise, negatively affecting the image fit of retrieved and generated images.

³⁹Please see Table 8 in the Appendix for a complete overview of all results.

Subtask 2 – Summary. The evaluation results show that both generative and retrieval methods perform well. Compared with subtask 1, we also see that scores across the board are slightly lower. This is not surprising, as random selection of items in subtask 2 is no longer allowed for manual adjustment or image checking. We also see that generative-based methods perform slightly better. Having in mind that retrieval-based methods are already more mature—meaning the teams have more experience in tuning and adapting the algorithms to specific scenarios—the generative-based methods seem to have a big potential to dominate in the image assignment tasks. Retrieval-based methods depend heavily on the underlying image dataset. The evaluation shows that a large dataset is essential for reaching good results.

When we analyze image variance, we find that the top-performing generated images have lower standard deviations (0.73–0.75 vs. 0.89–0.91), indicating greater consistency. This is because the generation process tailors images to each article, whereas the retrieval process is constrained by database availability [42], and some users dislike certain stock images [25]. The generation approach’s relative success suggests that flexibility in creating symbolic or illustrative imagery matters more than photographic authenticity for news thumbnails [42]. These findings highlight that retrieval remains the more reliable method for news-image recommendation due to its semantic precision, whereas generative approaches such as SDXL show potential to enhance visual diversity (e.g., using different art styles better match the mood of an article) [39].

4.4 Dataset Overview

All image recommendations and user ratings are available online as part of an open-source dataset for training, testing, and evaluation.⁴⁰ This includes all 22 retrieval and 16 generative workflows listed in Section 3, the editorial baseline, and a basic retrieval workflow. For subtask 1, there are 40 image recommendations (incl. one set of baseline images) for each of the 30 news headlines.⁴¹ For subtask 2, there are 27 image recommendations (incl. one set of baseline images) for each of the 20 news headlines. These recommendations are all grouped by article ID. Complementary to the recommendations, there are 52,200 user ratings from the online evaluation (on a 5-point Likert scale, with 30 ratings for each of the 1,740 image recommendations across the two subtasks).

Furthermore, we provide an alternative version of the dataset that is grouped by method.⁴² For subtask 1, this includes the same image recommendations that are included in the version that is grouped by article ID.⁴³ For subtask 2, *all* image recommendations produced by the 26 methods are included. Each method produced up to 8,500 recommendations,⁴⁴ for a total of 210,000 image recommendations. All images are provided as PNG files with dimensions of 460x260 pixels.⁴⁵

5 Lessons Learned

Following the analysis of the results, we summarize the insights as *Lessons Learned* from the online evaluation to provide a guideline for designing improved recommendation workflows. In particular, we would recommend and focus on ethical considerations, building hybrid workflows, using multi-stage extensions and multi-modal workflows, prompt engineering, offline evaluation for testing, and stylistic preferences.

⁴⁰<https://github.com/Informfully/Challenges/tree/main/newsimages25/images>

⁴¹There are four recommendations that consist of an empty placeholder. This is because the underlying method returned no image recommendations.

⁴²<https://seafle.ifi.uzh.ch/d/1f14d6d4306340e082d6/>.

⁴³The editorial baseline is removed from this part of the dataset. Furthermore, the placeholder images for the invalid image submissions were removed.

⁴⁴Six methods yielded incomplete results. These are marked accordingly.

⁴⁵For more technical specifications, please look at the online README: <https://github.com/Informfully/Challenges/blob/main/newsimages25/README.md>

Ethical Guidelines and the Role of Human Supervision. The experiment presented in this paper includes a subtask on human-AI collaboration and another subtask on fully automated recommendation workflows. In this context, however, it is important to note that all proposed methods are used exclusively under human supervision. Both retrieval and generation tasks recommend items with high variance in fit and overall image quality.

In a production environment, this necessitates oversight from editors, journalists, and the newsroom. Doing so ensures a consistent level of quality. The proposed techniques complement their work. Furthermore, human oversight enables the critical assessment of candidate images and ensures that strict editorial guidelines are followed when assigning news visuals; the proposed methods are currently not sensitive to ethical considerations.

To facilitate this oversight, it is crucial that any recommendation workflow be transparent, explainable, and easy to override. One element that would help enable this is a post-processing step that does automated content filtering. This could be an effective way to flag potentially harmful content and present it to editors for manual verification and approval.

Combining Retrieval and Generation Methods for Hybrid Workflows. A detailed analysis of all evaluated image recommendations showed that there are article- and category-specific properties that favor one approach over the other. For example, image retrieval achieved the best results for political news by recommending a picture of the particular person mentioned in the headline. And generative approaches received the highest ratings for articles about more abstract topics, such as economic news about inflation. Table 6 provides an overview of what article categories worked best with what type of recommendation workflow.

News that requires a high level of trust, such as breaking news, politics, or crime, needs factual grounding provided by photos taken during the event in question [3, 66]. Recency tends to play an important role. These items usually have a short shelf life. This makes retrieval systems the best option. In contrast to that, news categories, such as culture, environment, technology, and lifestyle, often make use of illustrations and could benefit from having access to more stylized images offered by a generative pipeline [3, 66]. Of course, there are exceptions in both cases. For example, political caricatures and cartoons could be easily generated. Travel reports as part of a lifestyle and culture article, on the other hand, should ideally use photos from the actual location rather than generated imagery. However, it is important to note that the classification presented here is but one possible way of categorizing articles.

Table 6. Overview of news categories, the typical format of visuals associated with them, and the recommendation technique that produced the best results.

News Category	Format	Recommended Technique
Latest News	Photos	Retrieval
Economy	Illustrations, Graphs	Generation
Politics	Photos, Caricatures	Retrieval, Generation
Crime	Photos	Retrieval
Sports	Photos	Retrieval
Culture & Travel	Photos, Illustrations	Retrieval, Generation
Environment	Illustrations	Generation
Technology	Photos, Illustrations	Retrieval, Generation
Lifestyle & Trends	Illustrations	Generation

Using a hybrid recommendation allows consideration of category-specific requirements. In this context, such a workflow incorporates both a retrieval and a generative pipeline to serve both types of recommendations. It can use an initial preprocessing step to determine the article type and then execute one of the two workflows. Alternatively, the workflow could run both retrieval and generation tasks in parallel, with a post-processing step at the end to determine which image recommendation best fits.

Multi-Stage Extensions and Multi-modal Workflows. The user study showed that several extensions to the basic workflow (cf. Figures 1 and 2) can significantly improve overall performance. The stage-specific techniques highlighted here include prompt augmentation (pre-processing stage), fine-tuning and ensembles (in-processing stage), and ranking/judging (post-processing stage).

Prompt augmentation uses techniques such as NER [61] or LLMs [33] to generate an input prompt for the retrieval/generation model. The most effective approach here is to reformulate the title into a complete sentence with negative prompts. This can also be used to verify that the input meets the requirements outlined in the respective model documentation.

Fine-tuning of models was shown to be among the most effective ways to achieve high article-image fit scores [34]. During fine-tuning, a news-specific dataset is used to update the original model weights to better fit the target domain. An alternative approach to changing the in-processing stage is to use ensembles, combining multiple models and configurations to produce a set of potential embeddings (for retrieval) or images (for generation). When taking this option, however, a post-processing step is needed to make a final selection from among all candidate items.

Judges and rankings are two instances of post-processing steps used to narrow down candidates. Using a VLM judge in fully automated settings [21] or a human judge in a collaborative approach [48] enables the comparison of multiple image recommendations. Alternatively, a ranking method can be introduced that assesses images against a predefined list of attributes and ranks them based on their overall fit with the news headline.

Here, multi-stage extensions can be complemented with multi-modal workflows. A multi-modal workflow uses both textual and visual prompts for retrieval and generation. For example, instead of starting with the news headline as input to a model, editors could first perform an archive search to narrow down the type of image they are looking for. Afterward, they can prompt the retrieval/generation model to recommend items similar to the results of their archival search. This approach presents a more effective control over the model's output.

Guidelines for Prompt Engineering. Results show that formulating the text input prompt for retrieval and generative models is one of the most important factors for producing good results. Being succinct is a key factor here. This means using a short title with precise terms and proper nouns is more effective than adding more article (meta)information, such as the lead or tags [34, 48]. At the same time, it is important that the prompt is semantically correct and remains a complete sentence (i.e., not too short, consisting only of one or two words [42, 66]).

While a richer input prompt with more details allows for capturing more information from the article itself, this procedure also introduces more “noise.” From the model's perspective, the additional information makes it more challenging to focus on the headline's critical parts [48]. The only way additional information positively affects overall performance is when it comes in the form of negative prompts, where they are an effective way to improve image selection [66].

Lastly, we saw that newer models tend to outperform older ones [34, 48]. Given the rapid pace of current development, even models that are less than half a year apart can make significant differences. When switching to a new model, however, it is always important to consult the official documentation for more technical details and the precise formulation of the input query.

Quality Assessment and the Limits of Offline Evaluation. Models for image retrieval and generation require large amounts of labeled data for training. Here, the main optimization criterion is similarity; the closer the project of a text and image embedding, the better. However, the analysis of the results shows that similarity-based offline measurement of models and workflows is not indicative of their online performance during the user study [33, 54].

The results showed that MMR, precision, recall, and nDCG, which are used to assess image-text similarity, can all yield predictions inconsistent with empirical results [39]. Furthermore, these measures may show significant differences in offline benchmarking between models, whereas human annotators perceive no differences in image quality [17].

As a result, it is critical not to rely solely on offline metrics to assess a solution's performance. Workflows need to be frequently tested with users and under "real" conditions (i.e., conditions similar to the settings of the final evaluation).

Accommodating Stylistic Preferences. Guidelines for prompting typically answer the question of *how* to prompt. They help you to reach your goal (e.g., a photorealistic depiction of a human) by answering the question of *how to formulate* the input query to get to the desired result. There are clear guidelines on which prompts work and which do not. But there exists an additional set of questions orthogonal to these more technical aspects. This is the set of stylistic questions regarding *what* to prompt. With the news, we need to ask ourselves *what image subject* [66] to focus on, *what art style* [48] to use, and *what mood/emotion* to convey [66] for framing a given article.

The analysis of the user study results clearly shows that image subject choices, e.g., using human-centric prompts to focus on individuals as image subjects [66], art style choices, e.g., photos vs. cartoons/caricatures [48], and selecting different emotional cues [66] can have a significant impact on the article rating. Unlike the technical aspects of prompting, there are no general rules available; ratings of different styles and subject choices are shaped by participants' cultural backgrounds, aesthetic preferences, and article category [48]. Therefore, finding a prompting technique that yields high-quality image recommendations requires empirical research examining readers' backgrounds and preferences across article categories.

Regardless of personal preferences, exploration remains critical. The prompts for both retrieval and generative methods should explicitly specify the desired subject, style, and mood, rather than leaving these unspecified. Instead of simply using the title, further instructions could be provided, e.g., selecting a picture with a happy mood, combined with a specific subject and art style. Get a sufficiently extensive coverage of all the options when testing.

6 Limitations and Future Work

A number of limitations apply to the proposed methods and the online survey. The two main issues we want to point out here focus on the evaluation procedure and the selection of the dataset and metrics. Afterward, we would like to highlight two potential future research directions. We propose looking beyond text-image similarity for thumbnail selection and addressing the challenges posed by misinformation in both image generation and retrieval.

Evaluation and Participant Demographics. During the evaluation, participants only looked at the headlines without the article body. The primary reasons for this were feasibility and limited time. In subsequent studies, the full article text should be shown (with fewer articles per person to avoid fatigue). Furthermore, the panel consisted only of experts, and the sample size was limited. Subsequent evaluations should include a broader selection of participants that more accurately represent the average readership and audience of a news outlet. Additionally, our survey interface lacks high external validity: it does not replicate the mobile-first environment, e.g., a news app on a phone, where most news consumption occurs these days.

Metrics and Dataset Constraints. We only look at image fit in the user study. However, several additional dimensions warrant consideration in subsequent evaluations. These metrics include, among others, visual quality, trustworthiness, and the absence of bias or stereotypes. Regarding retrieval, the weaker performance of the retrieval approaches may be in part due to a possible database mismatch. YFCC100M contains primarily travel and lifestyle photography rather than news-relevant imagery. For abstract concepts (economic policy, diplomatic tensions), semantically appropriate photographs simply do not exist, forcing CLIP to select weakly related images [42].

Beyond Text-Image Similarity. The majority of the literature on image retrieval and generation focuses on text-image similarity to build and assess the performance of a solution [53]. Similarity might be easy to operationalize for offline testing. When it comes to online image assessment for news thumbnails, this practice does not adequately reflect what people are looking for in an image (cf. divergent results between offline assessments and online user ratings [33, 54]). In our evaluation, we found that participants also assessed fit in terms of visual quality and the selection of art style. Similar to the previous point about using more metrics in online evaluation, we propose that in-production systems use more metrics to design and build retrieval/generation pipelines from the outset, e.g., surprising or novel elements to catch the reader’s attention. Ideally, this is complemented by frequent online user testing to ensure a consistent level of quality.

Ethical Guidelines and Misinformation. There are several ongoing debates about the role of visual content, such as photographs, videos, or other illustrations, in spreading or promoting misinformation [43]. A large number of news outlets have already published guidelines for the use of generative AI or have completely banned it on their platforms. Policies prohibits AI for stories/visuals by the *Associated Press*,⁴⁶ list human-in-the-loop requirement by *The New York Times*,⁴⁷ need senior editor approval by *The Guardian*,⁴⁸ limit AI for translation/subtitles by the *BBC*,⁴⁹ or exclude AI-written text by *Wired*.⁵⁰ Both subtasks followed existing media guidelines to prevent the promotion of misinformation. For image generation, we emphasized non-photorealistic images for article submissions to avoid images that could be easily mistaken for real-life events.⁵¹

7 Conclusion

Thumbnails for news articles play an important role in predicting user engagement. While they are critical in capturing the user’s attention, their impact is typically overlooked in RS evaluation. We address this research gap by contributing an open source dataset with over 210,000 image recommendations for 8,500 news articles, along with user ratings for training and evaluating future models. In our paper, we outline and assess all 39 retrieval and generation pipelines that comprise this dataset. We empirically validated all approaches and have users assess the article-image fit. These labels can be used to train and refine subsequent image retrieval and generation approaches. Furthermore, the online experiment allowed us to record user preferences and showed that the current focus on text-image similarity is insufficient to adequately address users’ expectations. Retrieval and generation models should, e.g., incorporate surprise or novelty alongside similarity when providing image recommendations. Finally, by providing a compilation of *Lessons Learned*, we share the main insight of our experiments to help with the development of upcoming systems.

⁴⁶<https://www.ap.org/the-definitive-source/behind-the-news/standards-around-generative-ai>

⁴⁷<https://www.nytc.com/press/principles-for-using-generative-ai-in-the-times-newsroom>

⁴⁸<https://www.theguardian.com/help/insideguardian/2026/mar/04/how-the-guardian-is-using-genai>

⁴⁹<https://www.bbc.co.uk/editorialguidelines/guidance/use-of-artificial-intelligence>

⁵⁰<https://www.wired.com/about/generative-ai-policy/>

⁵¹For details on the guidelines considered for the subtasks, we refer to the AI-CODE Deliverable D3.1 on user-centered requirements definition (AI-CODE Consortium, 2025).

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A Appendix

A.1 Full Evaluation Subtask 1

Table 7. All submitted runs for Sub task 1 (small, “human in the loop”) sorted by the obtained score. The ID uniquely identified the runs. Each run is listed with the name of the team that created it, together with the original description used in the respective paper. Type GEN marks the generative-based algorithms; RET the retrieval-based algorithms.

Rank	ID	Team	Original Description	Type	Score
1	1-12	CVG-IBA	CVG-IBA_VIVID	GEN	3.401
2	1-09	CVG-IBA	CVG-IBA_PromptForge	GEN	3.233
3	1-34	SELab-HCMUS	HCMUSSELab_HumanSelect	GEN	3.115
4	1-22	Headline Hunters	Headline_Hunters_Blip	RET	3.096
5	1-11	CVG-IBA	CVG-IBA_TRACE	RET	3.080
6	1-10	CVG-IBA	CVG-IBA_SEEK	RET	3.074
-	1-30	BASELINE	Original Images	-	3.041
7	1-38	SELab-HCMUS	HCMUSSELab_VanilaTitleRealistic	GEN	3.019
8	1-14	Das-RU	Das-RU_reg	RET	3.014
9	1-31	AFourP	AFourP_OPENCLIP_CLIPRERANK	RET	3.008
10	1-13	Das-RU	Das-RU_longtext	RET	3.006
11	1-23	Headline Hunters	Headline_Hunters_OpenClipExhaustive	RET	2.999
12	1-37	SELab-HCMUS	HCMUSSELab_VanilaTitleCartoon	GEN	2.985
13	1-36	SELab-HCMUS	HCMUSSELab_LLMCriticTitle	GEN	2.985
14	1-17	Das-RU	Das-RU_SDXLREF	GEN	2.984
15	1-15	Das-RU	Das-RU_SDXL	GEN	2.971
16	1-16	Das-RU	Das-RU_SDXLNEG	GEN	2.968
17	1-24	Headline Hunters	Headline_Hunters_OpenClipSelective	RET	2.965
18	1-19	ELITE_CODERS	ELITE_CODERS_CLIP	RET	2.964
19	1-33	SELab-HCMUS	HCMUSSELab_GASumRealistic	GEN	2.940
20	1-21	ELITE_CODERS	ELITE_CODERS_STABLE_DIFFUSION	GEN	2.896
21	1-06	CERTH_ITI	CERTH-ITI_RUN_6	RET	2.893
22	1-32	SELab-HCMUS	HCMUSSELab_GASumCartoon	GEN	2.874
23	1-05	CERTH_ITI	CERTH-ITI_RUN_5	RET	2.861
24	1-35	SELab-HCMUS	HCMUSSELab_LLMCritic	GEN	2.858
25	1-01	CERTH_ITI	CERTH-ITI_RUN_1	RET	2.857
26	1-27	DACS-UM-RTL	um-rtl_clip-vit-large-vlm_judge	RET	2.812
27	1-39	VisionX	VisionX_CLIP	RET	2.741
28	1-26	DACS-UM-RTL	um-rtl_clip-vit-large-llm_rewriting	RET	2.712
29	1-18	Das-RU	Das-RU_sec	RET	2.712
30	1-04	CERTH_ITI	CERTH-ITI_RUN_4	RET	2.709
31	1-02	CERTH_ITI	CERTH-ITI_RUN_2	RET	2.703
32	1-28	SSN-CSE	SSN-CSE_CLIP_KEYBERT_YFCC	RET	2.677
33	1-08	CodingSoft	CodingSoft_SDTURBO	GEN	2.666
34	1-25	DACS-UM-RTL	um-rtl_clip-vit-large-aesthetics	RET	2.650
35	1-03	CERTH_ITI	CERTH-ITI_RUN_3	RET	2.645
36	1-20	ELITE_CODERS	ELITE_CODERS_SDXL	GEN	2.634
37	1-40	VisionX	VisionX_SD	GEN	2.607
38	1-29	SyntaxError404	SyntaxError404_Retrieval (*)	RET	2.593
39	1-07	CodingSoft	CodingSoft_CLIP	RET	1.898

A.2 Full Evaluation Subtask 2

Table 8. All runs for Task 2 (large, “fully automated”) sorted by the obtained score. The ID uniquely identified the runs. Each run is listed with the name of the team that created it, together with the original description used in the respective paper. Type GEN marks the generative-based algorithms; RET the retrieval-based algorithms.

Rank	ID	Team	Description	Type	Score
1	2-04	CVG-IBA	CVG-IBA_PromptForge	GEN	3.194
2	2-05	CVG-IBA	CVG-IBA_TRACE	RET	3.114
3	2-17	Headline Hunters	Headline_Hunters_Blip	RET	3.088
4	2-06	Das-RU	Das-RU_SDXL	GEN	3.031
5	2-08	Das-RU	Das-RU_SDXLREF	GEN	3.024
6	2-14	SELab-HCMUS	HCMUSSELab_LLMCriticTitle	GEN	2.983
-	2-22	BASELINE	Original Images	-	2.956
7	2-25	Das-RU	Das-RU_longtext	RET	2.953
8	2-07	Das-RU	Das-RU_SDXLNEG	GEN	2.938
9	2-15	SELab-HCMUS	HCMUSSELab_VanilaTitleCartoon	GEN	2.927
10	2-13	SELab-HCMUS	HCMUSSELab_LLMCritic	GEN	2.923
11	2-26	Das-RU	Das-RU_reg	RET	2.916
12	2-01	AFourP	AFourP_OPENCLIP_CLIPRERANK	RET	2.902
13	2-27	Das-RU	Das-RU_sec	RET	2.893
14	2-19	Headline Hunters	Headline_Hunters_OpenClipSelective	RET	2.890
15	2-12	SELab-HCMUS	HCMUSSELab_GASumRealistic	GEN	2.861
16	2-18	Headline Hunters	Headline_Hunters_OpenClipExhaustive	RET	2.860
17	2-16	SELab-HCMUS	HCMUSSELab_VanilaTitleRealistic	GEN	2.850
18	2-11	SELab-HCMUS	HCMUSSELab_GASumCartoon	RET	2.823
19	2-10	ELITE_CODERS	ELITE_CODERS_SDXL	GEN	2.745
20	2-03	CodingSoft	CodingSoft_SDTURBO	GEN	2.684
21	2-20	SSN-CSE	SSN-CSE_CLIP_KEYBERT_YFCC	RET	2.673
22	2-24	VisionX	VisionX_CLIP	RET	2.612
23	2-21	SyntaxError404	SyntaxError404_Retrieval	RET	2.554
24	2-09	ELITE_CODERS	ELITE_CODERS_CLIP	RET	2.443
25	2-02	CodingSoft	CodingSoft_CLIP	RET	1.933
26	2-23	VisionX	VisionX_SD	GEN	1.717

A.3 Online Survey

Survey Completion

0%100%

The Original 'Wilhelm Scream' Was Found, And It's A Call for Better Sound Effect Preservation (Article ID: 1559)

Please rate how fitting the image is for the given article title.

Very Poor Fit

Below Average Fit

Average Fit

Above Average Fit

Very Good Fit



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Very Poor Fit

Below Average Fit

Average Fit

Above Average Fit

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Very Poor Fit

Below Average Fit

Average Fit

Above Average Fit

Very Good Fit

Fig. 3. Screenshot from the online survey of one article title and the corresponding image recommendations. Participants rated up to 40 images per title on a 5-point Likert scale.

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